



NUMERICAL ANALYSIS

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Newton-Raphson Method

It is based on linearization of the nonlinear continuous function $f(x)$. That is, the zero of $f(x)$ is approximated by the zero of the tangent line of f(x).

Graphical Derivation of Newton-Raphson Method

Assume we have the nonlinear continuous function $y = f(x) = 0$ shown in Fig.1 and it is required to find its root (r). Let xo be the initial estimate of r.

$$\tan\theta = \frac{dy}{dx}|_{(x=x_0)} = f'(x = x_0) = \frac{f(x_0)}{x_0 - x_1}$$

Solve for x_1

$$\boldsymbol{x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}}$$

Follow the same procedure to find x_2

$$\boldsymbol{x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}}$$

In general use the first approximation to get a second, the second to get a third, and so on, using the following numerical scheme

$$\boldsymbol{x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \qquad i = 0, 1, 2, \ldots \qquad f'(x_i) \neq 0} \qquad \textbf{(1)}$$

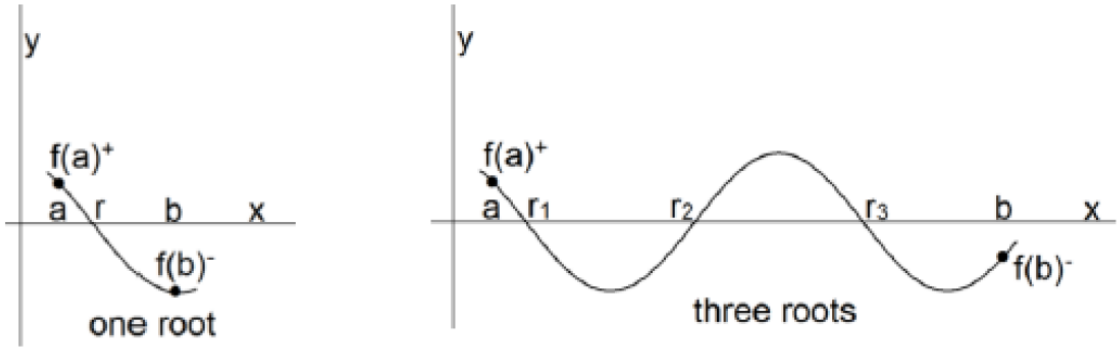
The stopping criterion is : $|x_{i+1} - x_i| \leq T_x$ where T_x is the tolerance for x.

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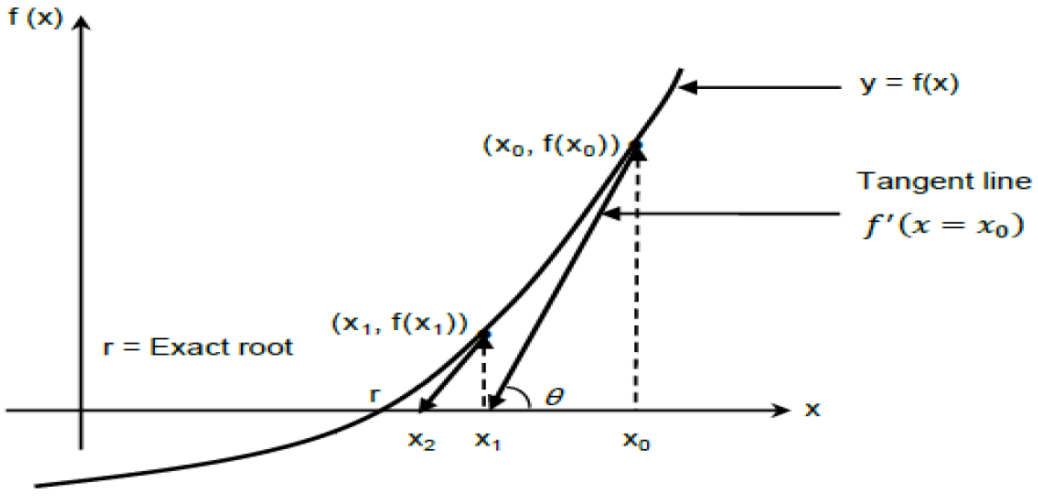
1. Open Methods: Sometimes we need to find the root of an equation near a point x. Use $\boldsymbol{x_0 = x}$.

2. Bracketing methods: Sometimes we need to find the root(s) of an equation in an interval $[\boldsymbol{a}, \boldsymbol{b}]$. Examine the sign of $\boldsymbol{f(x)}$ at the ends of the interval $[\boldsymbol{a}, \boldsymbol{b}]$. There are two cases:

• If $\boldsymbol{f(a)\, f(b) < 0}$, then there is one root or odd number of roots.

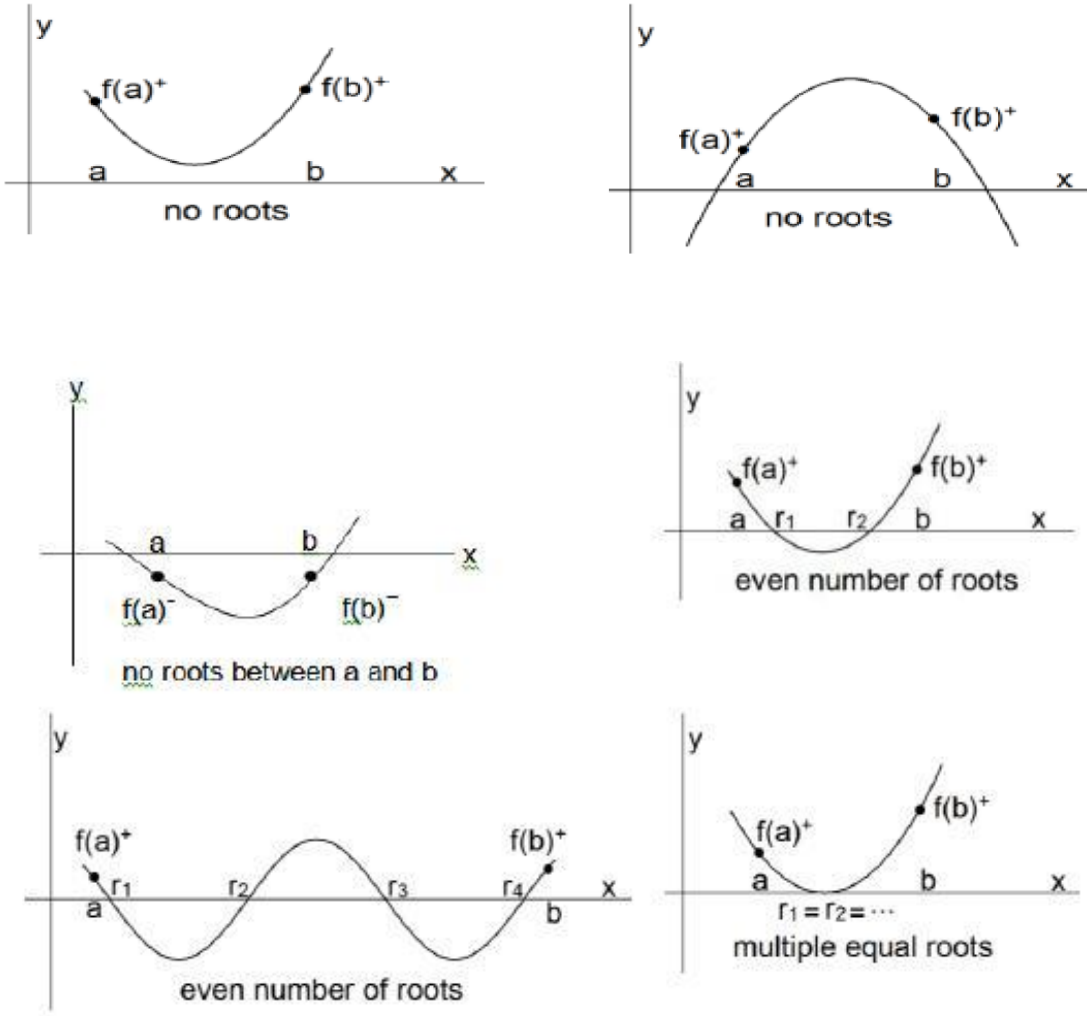


• If $\boldsymbol{f(a)\, f(b) > 0}$, then there are no roots, even number of roots, or multiple equal roots.



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Figure 1 : Geometrical illustration of Newton-Raphson method



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Example 1 : Use NR method to find the real root of $x^2 - x - 1 = 0$ correct to 5 decimal places (dp) in the interval $[-4,4]$. Choose $\Delta x = 1$.

Solution

Determine the position of the root

x	-4	-3	-2	-1	0	1		2	3	4
$f(x) = x^3 - x - 1$	-	-	-	-	-	-	$\downarrow -$	+	+	+

Determine the initial point x_0. The best initial point is the point that makes the value of $f(x)$ closer to zero.

x	$1(x_1)$	$2(x_2)$	$1.5(x_3 = (x_1 + x_2)/2)$
$f(x) = x^3 - x - 1$	-1	5	0.875

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} = x_i - \frac{(x_i)^3 - x_i - 1}{3(x_i)^2 - 1}$$

i	$x_i(x\ old)$	$x_{i+1}\ (x\ new)$	$E_i = \lvert x_{i+1} - x_i \rvert$	Notes
0	$1.50000(x_0)$	$1.34783(x_1)$	$E_0 = 0.15217 > T_x$	-
1	$1.34783(x_1)$	$1.32520(x_2)$	$E_1 = 0.02263 > T_x$	$E_1 < E_0\ ok$
2	$1.32520(x_2)$	$1.32472(x_3)$	$E_2 = 0.00048 > T_x$	$E_2 < E_1\ ok$
3	$1.32472(x_3)$	$1.32472(x_4)$	$E_3 = 0.00000 = T_x$	$E_3 < E_2\ ok$

The root is $r = 1.32472$ and $f_{(x=1.32472)} = 0.00001\ to\ 5dp$

Example 2 : Use NR method to estimate the positive **abscissa** (x-coordinate) of the intersection point of $f_1(x) = \sin x$and $f_2(x) = x^2$ correct to 4dp.

Solution :

The intersection point (points) of $f_1(x)$ and $f_2(x)$

x	0	0.5		1
$f(x) = \sin x - x^2$	0	+	$\downarrow r$	-

Determine the initial point x_0. The best initial point is the point that makes the value of $f(x)$ closer to zero.

x	$0.5(x_1)$	$1(x_2)$	$0.75(x_3 = (x_1 + x_2)/2)$
$f(x) = \sin x - x^2$	0.229	-0.159	0.119

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} = x_i - \frac{\sin x_i - x_i^2}{\cos x_i - 2x_i}$$

i	$x_i(x\ old)$	$x_{i+1}\ (x\ new)$	$E_i = \lvert x_{i+1} - x_i \rvert$	Notes
0	$0.7500(x_0)$	$0.9051(x_1)$	$E_0 = 0.1551$	-
1	$0.9051(x_1)$	$0.8777(x_2)$	$E_1 = 0.0274$	$E_1 < E_0\ ok$
2	$0.8777(x_2)$	$0.8767(x_3)$	$E_2 = 0.0010$	$E_2 < E_1\ ok$
3	$0.8767(x_3)$	$0.08767(x_4)$	$E_2 = 0.0000$	$E_3 < E_2\ ok$